## 8.3 Trigonometric Integrals

Trigonometric integrals involve algebraic combinations of the six basic trigonometric functions.

1. Basic formulas
2. Basic substitutions
3. Integration by parts
4. Use Identities
5. Multiplying by a form of 1
6. Eliminating square roots
7. Products of powers of sines and cosines
8. Integrals of powers of $\tan x$, $\sec x$, $\csc x$, and $\cot x$
9. Products of sines and cosines (multiple angles)

## Products of Powers of Sines and Cosines

Integrals of the form $\int \sin^m x \cos^n x \, dx$, where $m$ and $n$ are nonnegative integers.

### Case 1: $m$ is odd or $n$ is odd

- **When $m$ is odd and $n$ is even**

  Write $\sin^m x = \sin^{m-1} x \sin x$.

  Use the identity of $\sin^2 x = 1 - \cos^2 x$.

  Then substitute $u = \cos x$.

- **When $m$ is even and $n$ is odd**

  Write $\cos^n x = \cos^{n-1} x \cos x$.

  Use the identity of $\cos^2 x = 1 - \sin^2 x$.

  Then substitute $u = \sin x$.

- **When $m$ and $n$ are both odd**

  Any of the above methods can be applied.

**Example:**

$$\int \sin^3 x \cos^2 x \, dx =$$

**Example:**

$$\int \cos^5 x \, dx =$$

**Example:**

$$\int \sin^7 x \cos^5 x \, dx =$$

**Case 2: Both $m$ and $n$ are even**

Use the following identities:

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

To reduce the integrand to one in lower powers of $\cos 2x$.

**Example:**

$$\int \sin^2 x \cos^4 x \, dx = \frac{1}{16}\left(x - \frac{1}{4}\sin 4x + \frac{1}{3}\sin^3 2x\right) + C$$

**Integrals of powers of** $\tan x$**,** $\sec x$**,** $\csc x$**, and** $\cot x$

Use the identities

$$1 + \tan^2 x = \sec^2 x \quad \& \quad 1 + \cot^2 x = \csc^2 x$$

to integrate by substitution or by parts when necessary.

**Example:**

$\int \tan^4 x \, dx =$

**Example:**

$$\int \sec^3 x \, dx =$$

**Products of Sines and Cosines** (multiple angles)

Integrals of the forms

$$\int \sin mx \cos nx \, dx$$

$$\int \sin mx \sin nx \, dx$$

$$\int \cos mx \cos nx \, dx$$

Use identities

$$\sin mx \sin nx = \frac{1}{2}[\cos(m - n)x - \cos(m + n)x]$$

$$\sin mx \cos nx = \frac{1}{2}[\sin(m - n)x + \sin(m + n)x]$$

$$\cos mx \cos nx = \frac{1}{2}[\cos(m - n)x + \cos(m + n)x]$$

**Example:**

$\int \sin 3x \cos 5x \, dx =$

**Example:**

$\int \cos 3x \cos 4x \, dx =$